1)

a) $x_1(t) = 3u(t-2)$, $x_2(t) = 2u(t+2)$, $y(t) = x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau)\, x_2(t-\tau)\, d\tau$

from this 2 information we obtain $y(t) = \int_{-\infty}^{\infty} 3u(\tau-2)\, 2u(t-\tau+3)\, d\tau$

$u(\tau-2)$ is 1 if $\tau \geq 2$ and zero otherwise.

Likewise $u(t-\tau+3)$ is 1 if $\tau \leq t+3$ and zero otherwise

The intersection of these two will be the limits of the integral $\longrightarrow$

$$y(t) = \int_{2}^{t+3} 3 \cdot 2\, d\tau = 6\tau \Big|_{2}^{t+3} = 6t + 18 - 12 = 6t+6$$

$\Rightarrow$

$$\boxed{y(t) = 6(t+1)u(t+1)}$$

b) $x_1(t) = e^{-at}u(t)$, $x_2(t) = e^{-bt}u(t)$, $y(t) = x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau)\, x_2(t-\tau)\, d\tau$

then we have $y(t) = \int_{-\infty}^{\infty} e^{-a\tau} u(\tau)\, e^{-b(t-\tau)} \cdot u(t-\tau)\, d\tau$

for $t < 0$, the overlap is zero, so $y(t) = 0$

for $t \geq 0$,

$$y(t) = \int_{0}^{t} e^{-a\tau} e^{-b(t-\tau)}\, d\tau = e^{-bt} \int_{0}^{t} e^{-a\tau + b\tau}\, d\tau$$

$$\Rightarrow \quad y(t) = e^{-bt} \left. \frac{e^{-a\tau + b\tau}}{-a+b} \right|_0^t = e^{-bt} \frac{\left(1 - e^{-at+bt}\right)}{a-b}$$

therefore $$x_1(t) * x_2(t) = \frac{e^{-bt} - e^{-at}}{a-b} u(t)$$

c) $$x(t) = \begin{cases} 1 - \frac{|t|}{T}, & |t| < T \\ 0, & \text{elsewhere} \end{cases}$$

same as before $y(t) = x(t) * x(t) = \int_{-\infty}^{\infty} x(\tau)\, x(t-\tau)\, d\tau$

$x(t)$ is an even function (symmetric around the y-axis)

therefore $y(t)$ will also be even. We only need to calculate it for $t \geq 0$ and add $|t|$ at the end

width of $x(t)$ is $2T$ (from $-T$ to T) so the width of the convolution will be the sum of the widths: $2T+2T=4T$

so $y(t)$ is nonzero from $-2T$ to $2T$

for $0 \leq t < T$

peaks of triangles are close. The integral involves three seperate sections of piecewise linear functions multiplying each other. Evaluating the integral:

$$y(t) = \frac{2T}{3} - \frac{t^2}{T} + \frac{t^3}{2T^2}$$

for $T \leq t \leq 2T$

The triangles are moving away from each other. The overlap is smaller. The integral simplifies to:

$$y(t) = \frac{(2T - t)^3}{6T^2}$$

for $t \geq 2T$

The triangles no longer touch:

$$y(t) = 0$$

→ conclusion: By convolving 20 or more such triangular pulses, we will obtain a Gaussian Curve, which is a direct application of the Central Limit Theorem. Since convolving signals is mathematically identical to summing independent random variables, repeating this process causes to converge toward a normal distribution.

2)

we know $X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$

a) $A\sin(\omega_0 t + \phi) = \frac{A e^{j(\omega_0 t + \phi)}}{2j} - \frac{A}{2j} e^{-j(\omega_0 t + \phi)}$

↑ Euler's identity

we will use $e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)$ transform pair

$$= \frac{A}{2j} e^{j\phi} 2\pi\delta(\omega - \omega_0) - \frac{A}{2j} e^{-j\phi} 2\pi\delta(\omega + \omega_0)$$

$$\overset{\uparrow}{=} -A\pi j e^{j\phi}\delta(\omega - \omega_0) + A\pi j e^{-j\phi}\delta(\omega + \omega_0)$$

$\frac{1}{j} = -j$

b) $x(t) = 5e^{-a|t|}$ and $a>0$

we will use $e^{-a|t|} \longleftrightarrow \frac{2a}{a^2+\omega^2}$

$$x(j\omega) = 5\,\frac{2a}{a^2+\omega^2} = \frac{10a}{a^2+\omega^2}$$

c) $x(t) = \underbrace{e^{-2|t-1|}}_{x_1}\ \underbrace{e^{r(t-2)}}_{x_2}$

$$\int_{-\infty}^{\infty} x(t)\,e^{-j\omega t}\,dt = \int_{\infty}^{1} e^{2t-2}\,e^{jt-2j}\,e^{-j\omega t}\,dt + \int_{1}^{\infty} e^{-2t+2}\,e^{jt-2j}\,e^{-j\omega t}\,dt$$

$$= e^{-2-2j}\int_{-\infty}^{1} e^{t(2+j-j\omega)}\,dt + e^{2-2j}\int_{1}^{\infty} e^{t(-2+j-j\omega)}\,dt$$

$$= e^{-2-2j}\left[\frac{e^{t(2+j-j\omega)}}{2+j-j\omega}\right]_{-\infty}^{1} + e^{2-2j}\left[\frac{e^{t(-2+j-j\omega)}}{-2+j-j\omega}\right]_{1}^{\infty}$$

... $= \dfrac{4\,e^{-j(\omega+1)}}{4+(1-\omega)^2}$

d) sin is periodic so we can say $x(t) = \frac{\sin 2\pi (t-2)}{\pi (t-2)}$

using $x(t - td) \xleftrightarrow{F} e^{-jwtd} G(jw)$

we can assume $G(jw) \begin{cases} 1, & W < w < W \\ 0, & \text{elsewhere} \end{cases}$

for $x_2(t) = \frac{\sin Wt}{\pi t}$

$$g(t) = \frac{1}{2\pi} \int_{-W}^{W} G(jw) e^{jwt} dw = \frac{1}{2\pi} \int_{-W}^{W} e^{jwt} dw = \ldots = \frac{e^{jWt} - e^{-jWt}}{2j\pi t}$$

$$= \frac{\sin(Wt)}{\pi t} = x_2(t)$$

$G(jw) = u(w+W) - u(w-W)$, $W = 2\pi$ since we have $\sin 2\pi$

$= u(w+2\pi) - u(w-2\pi)$

$$x(jw) = e^{-jw2} \cdot (u(w+2\pi) - u(w-2\pi)) = \begin{cases} e^{-2jw}, & -2\pi < w < 2\pi \\ 0, & \text{elsewhere} \end{cases}$$

$x(t) = \text{sinc}(at + b)$ where a and b are const $\text{sinc}(\theta) = \frac{\sin(\pi\theta)}{\pi\theta}$

same as before $X(j\omega) = u(\omega+\pi) - u(\omega-\pi)$

we will use $x(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|} X(j\omega/a)$ (scaling prop.)

we have $X(j\omega/a) = \begin{cases} 1, & -|a|\pi < \omega < |a|\pi \\ 0, & \text{elsewhere} \end{cases}$

so $x(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|}\left(u(\omega+\pi|a|) - u(\omega-\pi|a|)\right)$

shift by $\frac{b}{a} \Rightarrow x\left(a\left(t+\frac{b}{a}\right)\right) \xleftrightarrow{\mathcal{F}} \frac{e^{j\omega\frac{b}{a}}}{|a|}\left[u(\omega+\pi|a|) - u(\omega-\pi|a|)\right]$

$x\left(a\left(t+\frac{b}{a}\right)\right) = \text{sinc}(at+b)$

$= \frac{e^{-j\omega\frac{b}{a}}}{|a|}$ for $-|a|\pi < \omega < |a|\pi$ (zero elsewhere)

3) input $x(t) = e^{-2t}\,u(t)$

output $y(t) = (e^{-t} - e^{-3t})\,u(t)$, $H(j\omega) = ?$, $h(t) = ?$

$$x(t) \xleftrightarrow{\mathcal{F}} x(j\omega) = \frac{1}{2+j\omega}$$

$$y(t) \xleftrightarrow{\mathcal{F}} Y(j\omega) = \frac{1}{1+j\omega} - \frac{1}{3+j\omega} = \frac{2}{(1+j\omega)(3+j\omega)}$$

$$\Rightarrow H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{2(2+j\omega)}{(1+j\omega)(3+j\omega)}$$

$$H(j\omega) = \int_{-\infty}^{\infty} h(t)\, e^{-j\omega t}\, dt$$, we are supposed to use partial fractions

since $H(j\omega)$ is complex

$$H(j\omega) = \frac{A}{1+j\omega} + \frac{B}{3+j\omega} = \frac{2(2+j\omega)}{(1+j\omega)(3+j\omega)}$$

we have (from nominators) : $A(3+j\omega) + B(1+j\omega) = 2(2+j\omega)$

we can use $A=1$ and $B=1$

so $$H(j\omega) = \frac{1}{1+j\omega} + \frac{1}{3+j\omega}$$

and $$h(t) = e^{-t}\,u(t) + e^{-3t}\,u(t)$$

magnitude:

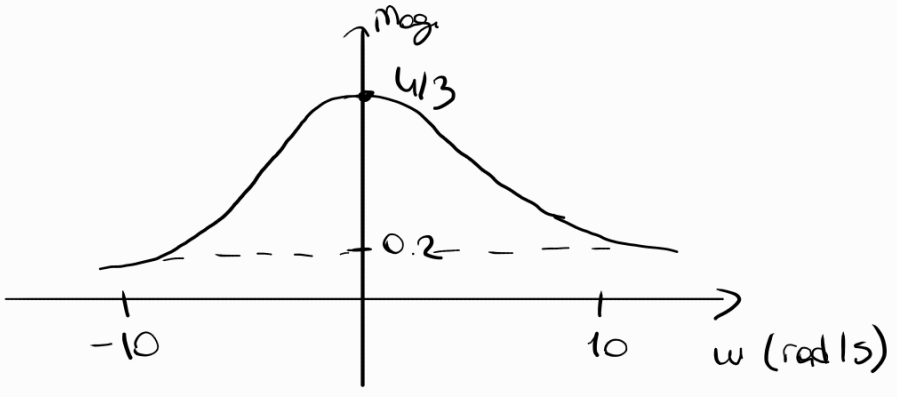
$$|H(j\omega)| = \frac{2\,|2+j\omega|}{|1+j\omega|\cdot|3+j\omega|} = \frac{2\sqrt{4+\omega^2}}{\sqrt{1+\omega^2}\sqrt{9+\omega^2}}$$

$$|a+j\omega| = \sqrt{a^2+\omega^2}$$

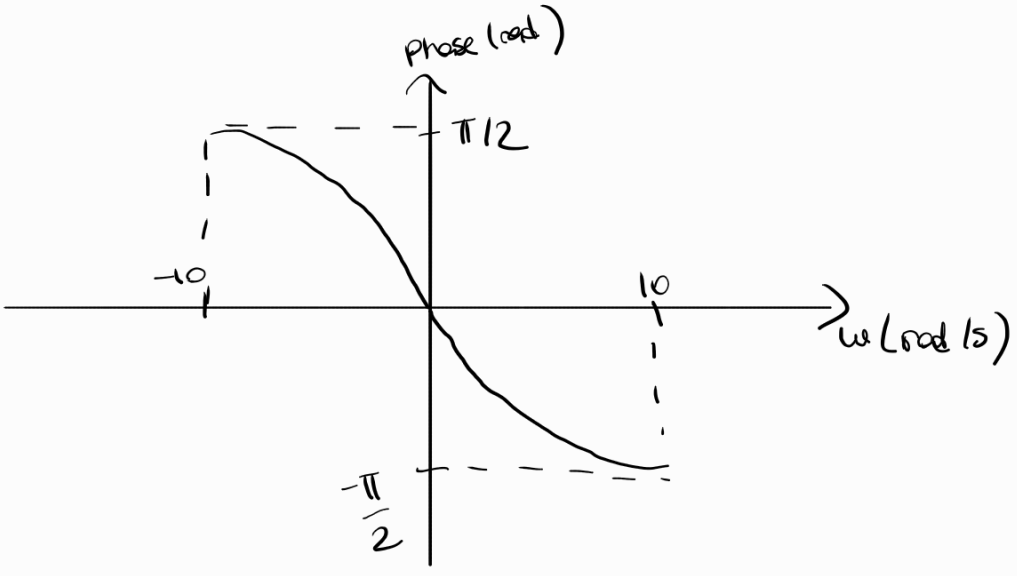
$$\angle H(j\omega) = \angle 2 + \angle 4+2j\omega - \angle 1+j\omega - \angle 3+j\omega$$

$$= \arctan\left(\frac{\omega}{2}\right) - \arctan(\omega) - \arctan\left(\frac{\omega}{3}\right)$$

Plots



Magnitude Plot



4)

a) $X(j\omega) = \pi e^{-2|\omega|}$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \underbrace{X(j\omega)}_{\pi e^{-2|\omega|}} e^{j\omega t}\, d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \pi e^{-2|\omega|} e^{j\omega t}\, d\omega$$

$$= \frac{1}{2}\left[\int_{-\infty}^{0} e^{2\omega} e^{j\omega t}\, d\omega + \int_{0}^{\infty} e^{-2\omega} e^{j\omega t}\, d\omega \right]$$

$$= \frac{1}{2}\left[\frac{e^{\omega(2+jt)}}{2+jt}\Big|_{-\infty}^{0} + \frac{e^{\omega(-2+jt)}}{-2+jt}\Big|_{0}^{\infty} \right]$$

$$= \frac{1}{2}\left(\frac{1}{2+jt} + \frac{1}{2-jt} \right) = \boxed{\frac{2}{4+t^2}}$$

b) $X(j\omega) = u(\omega-10) - u(\omega-20)$

$$x(t) = \frac{1}{2\pi} \int_{10}^{20} e^{j\omega t}\, d\omega = \frac{1}{2\pi} \frac{e^{j\omega t}}{jt}\Big|_{10}^{20}$$

$$= \frac{e^{15jt}}{\pi t} \underbrace{\frac{e^{5jt} - e^{-5jt}}{2j}}_{\sin 5t} \Rightarrow \frac{e^{15jt}}{\pi t} \sin 5t = x(t)$$

5)

$$x(t) = \sin(100\pi t) + 2\sin(200\pi t)$$

$$g(t) = x(t)\sin(500\pi t)$$

$$r(t) = g(t)\sin(500\pi t)$$

$$r(t) = x(t)\sin^2(500\pi t) =$$

$$r(t) = \frac{x(t) - x(t)\cos(1000\pi t)}{2} \qquad \left(\sin^2 500\pi t = \frac{1-\cos(1000\pi t)}{2}\right)$$

$$\Rightarrow r(t) = \underbrace{\frac{\left(\sin(100\pi t) + 2\sin(200\pi t)\right)}{2}}_{A} - \underbrace{\frac{\left(\sin(100\pi t) + 2\sin(200\pi t)\right)\cos(1000\pi t)}{2}}_{B}$$

terms of $x(t)$ has 100π and 200π which are smaller than 500π so they pass the filter ($\omega = 500\pi$ rad/s for the filter)

that's why A passes the filter. we will see B now:

$$B = -\frac{\sin(100\pi t)\cos(1000\pi t)}{2} - \frac{2\sin(200\pi t)\cos(1000\pi t)}{2}$$

$$B = -\frac{1}{4}\left[\sin(1100\pi t) - \sin(900\pi t) - \sin(1200\pi t) + \sin(800\pi t)\right]$$

$\omega > 500\pi$ for all, they don't pass the filter

therefore we have

$$y(t) = A = \frac{\sin(100\pi t)}{2} + \sin(200\pi t)$$